

Skills Summary

Equal Groups — How Many Groups? How Many in Each?

Use this reference while completing your worksheet or when studying.

Skill 1 — How many are in each group?

The three numbers in every sharing story:

- **total** — how many there are altogether (the biggest number)
- **groups** — how many bags, plates, pots, rows
- **in each** — how many go in one bag, one plate, one pot

When the story gives the number of groups: $\text{total} \div \text{groups} = \text{how many in each group}$.

Picture it: (• • •) (• • •) (• • •) — three groups, and three dots *in each*.

Example: 27 seeds are shared into 3 pots. How many seeds are in each pot?

Step 1. The story gives how many pots there are, so 3 is a group count and the missing number is the size of one group — share the total out.

Step 2. $27 \div 3$: skip-count by threes to 27, which takes nine of them. Each pot gets **9** seeds. Check: $3 \times 9 = 27$ ✓

Try it: 21 apples go into 3 baskets, the same number in each. How many apples are in each basket?

check: 7

Skill 2 — How many groups?

When the story gives how many go in each group: $\text{total} \div \text{in each} = \text{number of groups}$.

The words “**in each**” are the giveaway. “5 cups” counts groups; “5 in each cup” tells the size of a group.

Never write back a number the story gave you. If the story says 5 go in each cup, then 5 is not the answer to “how many cups?”

Example: There are 15 counters. Each cup holds 5 counters. How many cups are filled?

Step 1. The words in each tell us 5 is the size of a group, so the missing number is how many groups the total makes.

Step 2. $15 \div 5$: take away five at a time — fifteen, ten, five, none left, so three cups were filled. There are **3** cups. Check: $3 \times 5 = 15$ ✓

Try it: There are 28 crayons and 7 crayons go in each box. How many boxes are filled?

check: 4

Skill 3 — Check by multiplying back

The check: groups $\times$ in each group = total.

If the multiplication does not land on the total, the two numbers are doing the wrong jobs — most often because a number from the story was written back as the answer.

Example: There are 32 marbles with 8 in each jar. Someone says there are 8 jars. Check it, and give the right number of jars.

Step 1. Test the claim by multiplying back before doing anything else: eight jars of eight would need sixty-four marbles, more than the total, so the claim is wrong.

Step 2. $32 \div 8$: count up by eights — eight, sixteen, twenty-four, thirty-two — which is four jars. There are 4 jars. Check: $4 \times 8 = 32$ ✓

Try it: There are 45 stickers with 9 in each row. How many rows are there? Check by multiplying back.

check: 5

(!) Watch out: the answer is never a number the story already gave you. If the question repeats one of the numbers you were told, read it again — you have probably answered the other question.